Answer **all** the questions in the spaces provided.

1 **(a)** A ball of mass 0.80 kg is released from rest in air from a tall building. The variation of the velocity v of the ball with time t is shown in Fig. 1.1.

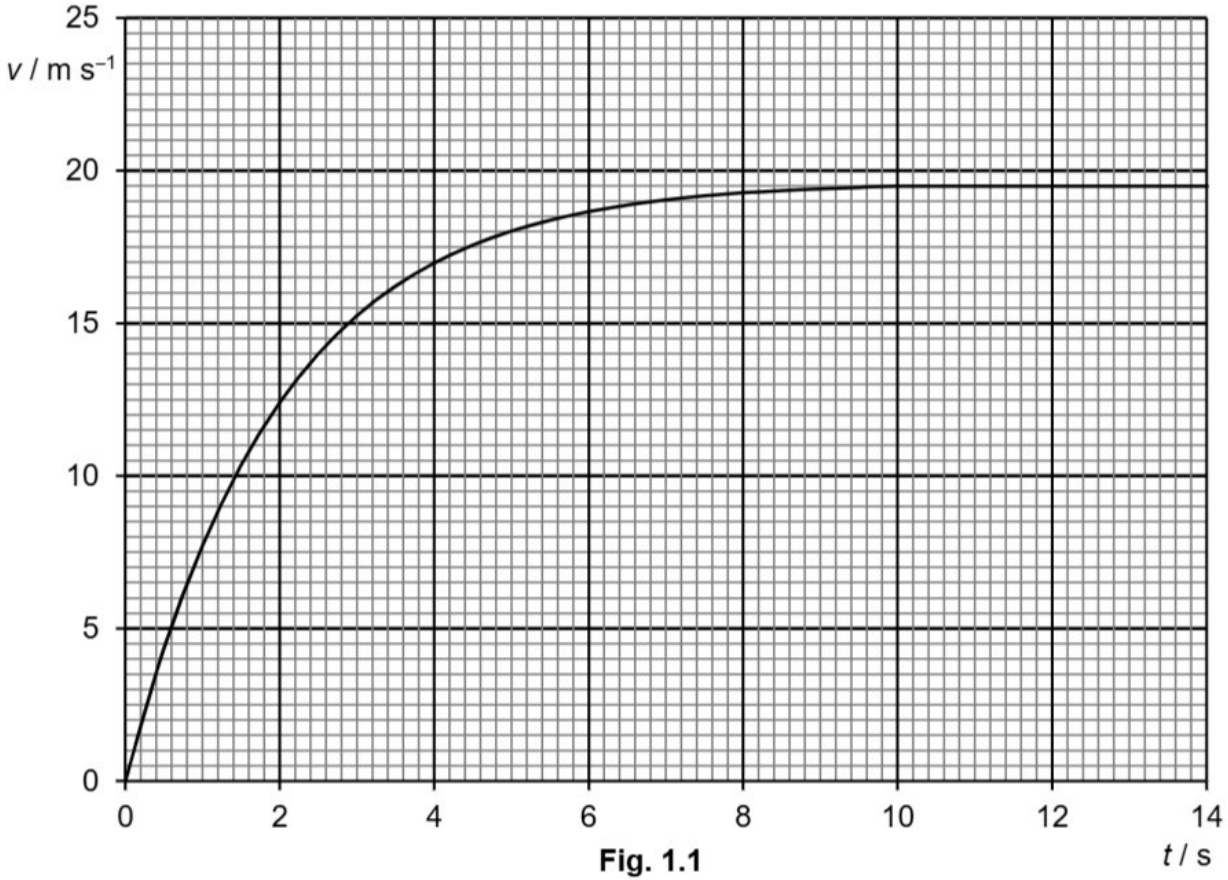


Fig. 1.1

(i) With reference to Fig. 1.1, explain how the drag force F_D acting on the ball changes as the velocity of the ball increases from $t = 0$ to $t = 8.0$ s.

.....

.....

.....

.....

..... [2]

(ii) Determine the magnitude of the drag force F_D acting on the ball at time t = 3.5 s.

F_D = ... N [3]

(b) The drag force F_D is given by the equation

$$F_D = 6\pi r \eta v$$

where r is the radius of the ball,
η is the viscosity of the air,
v is the velocity of the ball.

(i) Determine the SI base units of η.

base units = ... [3]

(ii) Explain how the maximum drag force will change if a ball of a greater mass, but of the same radius, is released from rest from the same building.

.....

.....

..... [1]

[Total: 9]